

Practical - 7

Aim:

Perform the file handling operations in Python.

Q1: Write a Python program to read content from a file and display the word with maximum length.

Code:

```
def max_len(li):  
    max = li[0]  
    for i in li:  
        if len(i) > len(max):  
            max = i  
    return max  
  
with open('/content/Practical-7/sample_text.txt') as file:  
    output = file.read()  
    max_len_string = max_len(output.split())  
    print("File Content: \n" + output)  
    print("\nWord With Max Length:", max_len_string)
```

Output:

```
File Content:  
Python is a powerful programming language  
ArtificialIntelligence is transforming industries  
Developers constantly improve their problemSolvingSkills  
Consistency and discipline create extraordinary results
```

```
Word With Max Length: ArtificialIntelligence
```

Q2: Write a Python program to combine each line from first file with the corresponding line in second file.

Code:

```
file1_lines = []
file2_lines = []
with open('/content/Practical-7/names.txt') as file:
    for line in file:
        file1_lines.append(line.strip())
with open('/content/Practical-7/scores.txt') as file:
    for line in file:
        file2_lines.append(line.strip())

output = []

for i in range(min(len(file1_lines), len(file2_lines))):
    output.append(file1_lines[i] + ' ' + file2_lines[i])

with open('/content/Practical-7/names.txt') as file:
    file_output = file.read()
    print("File1 Content: \n" + file_output)

with open('/content/Practical-7/scores.txt') as file:
    file_output = file.read()
    print("\nFile2 Content: \n" + file_output)

print("\nJoined Content:", output)
```

Output:

File1 Content:

Vatsal

Rahul

Anjali

Meera

Honey

File2 Content:

85

92

78

88

90

Joined Content: ['Vatsal 85', 'Rahul 92', 'Anjali 78', 'Meera 88',
'Honey 90']

Q3: Write a Python program to read first n lines of a file.

Code:

```
lines = int(input("How Many Line Do You Want to Read?: "))
with open('/content/Practical-7/long_text.txt') as file:
    for i in range(lines):
        output_lines = file.readline().strip()
        print(output_lines)
```

Output:

```
How Many Line Do You Want to Read?: 5
Line 1: Introduction to Python
Line 2: Variables and Data Types
Line 3: Conditional Statements
Line 4: Loops in Python
Line 5: Functions
```

Q4: Write a Python program to write students information into csv file and read that details.

Code:

```
csv_format = ('RollNo','Name','Age','Department','Marks')

student_info = [(101,'Vatsal',20,'Computer',88),
                (102,'Rahul',21,'Mechanical',75),
                (103,'Anjali',19,'Electrical',91),
                (104,'Meera',20,'Civil',84),
                (105,'Honey',22,'IT',79),]

with open('/content/sample_data/students_info.txt', 'w') as file:
    file.write(f"{','.join(csv_format)}\n")
    for i in student_info:
        file.write(f"{i[0]},{i[1]},{i[2]},{i[3]},{i[4]}\n")

stu_info = []

with open('/content/sample_data/students_info.txt') as file:
    csv_format_from_file = file.readline()
    for line in file:
        stu_info.append(line.strip().split(','))

print(stu_info)
```

Output:

```
[['101', 'Vatsal', '20', 'Computer', '88'],
 ['102', 'Rahul', '21', 'Mechanical', '75'],
 ['103', 'Anjali', '19', 'Electrical', '91'],
 ['104', 'Meera', '20', 'Civil', '84'],
 ['105', 'Honey', '22', 'IT', '79']]
```

Practice Exercise:**1. Write a program that performs text file analysis:**

- Read a text file containing student essays
- Create a new file that stores:
 - Total number of sentences
 - Average words per sentence
 - List of all words longer than 10 characters
- Output should be saved in 'analysis.txt'

Code:

```
with open('/content/sample_data/essays.txt') as file:
    essay = file.read()

# Remove extra whitespace
essay = essay.strip()

# Split into sentences and remove empty ones
sentences = [s.strip() for s in essay.split('.') if s.strip()]

num_of_sentences = len(sentences)

total_words = 0
word_longer_than_10_ch = []

for sentence in sentences:
    words = sentence.split()
    total_words += len(words)

    for word in words:
        clean_word = word.strip(",!?\"'")
        if len(clean_word) > 10:
            word_longer_than_10_ch.append(clean_word)

avg_words_per_sentence = total_words / num_of_sentences

analysis = f"""
Total number of sentences: {num_of_sentences}
Average words per sentence: {avg_words_per_sentence}
Words longer than 10 characters: {word_longer_than_10_ch}
"""

with open('/content/sample_data/analysis.txt', 'w') as file:
    file.write(analysis.strip())
print("Analysis is Saved in analysis.txt")
```

Output:

Analysis is Saved in analysis.txt

analysis.txt:

Total number of sentences: 36

Average words per sentence: 9.638888888888889

Words longer than 10 characters: ['Consistency', 'consistently', 'circumstances', 'consistency', 'predictable', 'consistently', 'improvements', 'significant', 'Consistency', 'disciplined', 'consistency', 'consistency', 'unproductive', 'intentional', 'distractions', 'entertainment', 'notifications', 'consistency', 'necessarily', 'consistency', 'extraordinary', 'inspiration']

2. Create a program to handle employee records:

- Write employee details (id, name, salary) to 'employees.csv'
- Read the CSV file and create two new files:
 - high_salary.txt (employees with salary > 50000)
 - low_salary.txt (employees with salary <= 50000)
- Each output file should maintain the same CSV format

Code:

```
with open('/content/sample_data/employees.csv') as file:
    csv_format = file.readline()
    high_sal = []
    low_sal = []
    for i in file:
        emp = i.strip('\n ').split(',')
        if int(emp[2]) > 50000:
            high_sal.append(emp)
        else:
            low_sal.append(emp)

with open('/content/sample_data/high_salary.csv', 'w') as file:
    file.write(csv_format)
    for i in high_sal:
        file.write(f"{','.join(i)}\n")

with open('/content/sample_data/low_salary.csv', 'w') as file:
    file.write(csv_format)
    for i in low_sal:
        file.write(f"{','.join(i)}\n")
```


employees.csv:

```
id,name,salary
101,Aarav Mehta,45000
102,Ishita Sharma,72000
103,Rohan Patel,50000
104,Ananya Verma,88000
105,Devansh Singh,32000
106,Meera Joshi,51000
107,Arjun Nair,49999
108,Kavya Iyer,120000
109,Vihaan Desai,54000
110,Sneha Kapoor,47000
111,Aditya Rao,60000
112,Priya Shah,30000
113,Karan Malhotra,50500
114,Nisha Reddy,48000
115,Ritik Gupta,150000
```

high_salary.csv:

```
id,name,salary
102,Ishita Sharma,72000
104,Ananya Verma,88000
106,Meera Joshi,51000
108,Kavya Iyer,120000
109,Vihaan Desai,54000
111,Aditya Rao,60000
113,Karan Malhotra,50500
115,Ritik Gupta,150000
```

low_salary.csv:

```
id,name,salary
101,Aarav Mehta,45000
103,Rohan Patel,50000
105,Devansh Singh,32000
107,Arjun Nair,49999
110,Sneha Kapoor,47000
112,Priya Shah,30000
114,Nisha Reddy,48000
```

3. Develop a program for log file processing:

- Read a log file containing timestamps and events
- Create summary files:
 - errors.txt (all error messages)
 - warnings.txt (all warning messages)
 - summary.txt (count of each type of event)
 - Original format: [timestamp] [type] [message]

Code:

```

errors = []
warnings = []
infos = []
with open('/content/sample_data/application.log') as file:
    for line in file:
        event_type = line.strip("\n ").split()[2]
        if event_type == '[INFO]':
            infos.append(line)
        elif event_type == '[WARNING]':
            warnings.append(line)
        elif event_type == '[ERROR]':
            errors.append(line)

with open('/content/sample_data/errors.txt', 'w') as file:
    for line in errors:
        file.write(line)
with open('/content/sample_data/warnings.txt', 'w') as file:
    for line in warnings:
        file.write(line)

count_errors = len(errors)
count_warnings = len(warnings)
count_infos = len(infos)

summary = f"""
Event Summary Report
-----
INFO: {count_infos}
WARNING: {count_warnings}
ERROR: {count_errors}
TOTAL EVENTS: {count_errors + count_warnings + count_infos}
"""

with open('/content/sample_data/summary.txt', 'w') as file:
    file.write(summary.strip())

```

application.log:

```
[2026-03-01 09:00:12] [INFO] Application started successfully
[2026-03-01 09:01:03] [INFO] Database connection established
[2026-03-01 09:02:45] [WARNING] High memory usage detected
[2026-03-01 09:03:10] [ERROR] Failed to load configuration file
[2026-03-01 09:04:22] [INFO] User login successful: user_id=101
[2026-03-01 09:05:55] [ERROR] Database timeout error
[2026-03-01 09:06:30] [WARNING] Disk space running low
[2026-03-01 09:07:18] [INFO] Scheduled backup completed
[2026-03-01 09:08:41] [ERROR] Invalid user credentials
[2026-03-01 09:09:05] [INFO] Password reset request processed
[2026-03-01 09:10:11] [WARNING] CPU usage above 85%
[2026-03-01 09:11:49] [ERROR] Failed to send email notification
[2026-03-01 09:12:33] [INFO] New user registered: user_id=145
[2026-03-01 09:13:27] [ERROR] Database timeout error
[2026-03-01 09:14:02] [INFO] Application shutdown initiated
[2026-03-01 09:14:45] [INFO] Application shutdown completed
```

errors.txt:

```
[2026-03-01 09:03:10] [ERROR] Failed to load configuration file
[2026-03-01 09:05:55] [ERROR] Database timeout error
[2026-03-01 09:08:41] [ERROR] Invalid user credentials
[2026-03-01 09:11:49] [ERROR] Failed to send email notification
[2026-03-01 09:13:27] [ERROR] Database timeout error
```

warnings.txt:

```
[2026-03-01 09:02:45] [WARNING] High memory usage detected
[2026-03-01 09:06:30] [WARNING] Disk space running low
[2026-03-01 09:10:11] [WARNING] CPU usage above 85%
```

summary.txt:

Event Summary Report

INFO: 8

WARNING: 3

ERROR: 5

TOTAL EVENTS: 16

4. Create a program for student attendance tracking:

- Read daily attendance files (date_attendance.csv)
- Calculate monthly attendance
- Generate report.csv with columns:
 - Student ID
 - Total Present Days
 - Total Absent Days
 - Attendance Percentage
 - Handle missing files and incomplete data

Code:

```
import glob

file_paths = glob.glob('/content/sample_data/Attendance/*')
student_attendance = {}

for path in file_paths:

    with open(path) as file:
        csv_format = file.readline()
        for line in file:
            student = line.strip().split(',')
            student_name = student[0]
            student_status = student[1]
            if student_name == '' or student_status == '':
                continue
            elif student_status == 'Present':
                if student_attendance.get(student_name) != None:
                    student_attendance[student_name]['Present'] += 1
                else:
                    student_attendance[student_name] = {'Present': 1,
'Absent': 0}
            elif student_status == 'Absent':
                if student_attendance.get(student_name) != None:
                    student_attendance[student_name]['Absent'] += 1
                else:
                    student_attendance[student_name] = {'Present': 0,
'Absent': 1}

print(student_attendance)

with open('/content/sample_data/report.txt', 'w') as file:
    # CSV Format
    file.write('StudentID,Total    Present    Days,Total    Absent
Days,Attendance Percentage\n')
    for student, attendance in student_attendance.items():
```

```

    # Calculate percentage
    percentage = attendance['Present'] / (attendance['Present'] +
attendance['Absent']) * 100

file.write(f"{student},{attendance['Present']},{attendance['Absent '
]},{percentage}\n")

```

2026-03-01_attendance.csv:

```

StudentID,Status
101,Present
102,Absent
103,Present
104,Present
105,Absent

```

2026-03-02_attendance.csv:

```

StudentID,Status
101,Present
102,Present
103,Present
104,Absent
105,Absent

```

2026-03-03_attendance.csv:

```

StudentID,Status
101,Absent
102,Present
103,Present
104,Present
105,Present

```

2026-03-05_attendance.csv:

```

StudentID,Status
101,Present
102,Present
103,
104,Present
,Absent
105,Absent

```

report.csv:

```

StudentID,Total Present Days,Total Absent Days,Attendance
Percentage
101,3,1,75.0
102,3,1,75.0
103,3,0,100.0
104,3,1,75.0
105,1,3,25.0

```